

FE 5209 Homework Assignment 1.

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1) Probability and Statistics

1.1) a) Given $x_1, \dots, x_n$ are i.i.d and

$$f(x) = \frac{1}{\theta} e^{-x/\theta} \text{ for } x > 0$$

Likelihood function. $L(\theta) = f(x_1, \dots, x_n | \theta)$ becomes

$$L(\theta) = f(x_1, \theta) f(x_2, \theta) \dots f(x_n, \theta) \text{ as}$$

$x_1, \dots, x_n$ are independent.

$$\Rightarrow L(\theta) = \prod_{i=1}^n \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^n} \cdot e^{-(\sum x_i)/\theta}$$

We get maximum likelihood estimator (MLE) $\hat{\theta}$ by maximizing $L(\theta)$.

$$\begin{aligned} \log L(\theta) &= \log \frac{1}{\theta^n} e^{-(\sum x_i)/\theta} \\ &= -n \log \theta - (\sum x_i)/\theta \end{aligned}$$

Taking derivative w.r.t θ and equating to 0 we get

$$\frac{d \log L(\theta)}{d\theta} = -\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} = 0$$

$$\hat{\theta} = \sum x_i / n = \bar{x}$$

The Maximum Likelihood Estimator MLE $\hat{\theta}$ of θ is $\bar{x}$ (Mean of $x_1, x_2, \dots, x_n$)

The mean $\bar{x}$ can be calculated by the following

$$E[x] = \int_0^{\infty} x \cdot f(x) dx \quad \text{as } x > 0$$

$$\begin{aligned}
&= \int_0^{\infty} x \cdot \frac{1}{\theta} e^{-x/\theta} dx \\
&= \left(\frac{x}{\theta} \int e^{-x/\theta} dx \right)_0^{\infty} - \frac{1}{\theta} \int_0^{\infty} \frac{dx}{dx} \int e^{-x/\theta} dx \\
&= \left(\frac{-x}{\theta} \cdot \theta \cdot e^{-x/\theta} \right)_0^{\infty} + \frac{1}{\theta} \int_0^{\infty} \theta \cdot e^{-x/\theta} dx \\
&= (-x e^{-x/\theta})_0^{\infty} - \theta (e^{-x/\theta})_0^{\infty} \\
&= -\lim_{x \rightarrow \infty} x e^{-x/\theta} + 0 - \theta(0-1)
\end{aligned}$$

$\lim_{x \rightarrow \infty} x e^{-x/\theta}$ can be found out by L'Hospital rule

$$= \lim_{x \rightarrow \infty} (dx/dx) / (d(e^{-x/\theta})/dx) = \lim_{x \rightarrow \infty} 1 / \theta e^{-x/\theta} \rightarrow 0$$

$$E[x] = 0 + \theta = \theta.$$

b) $\hat{\theta}$ is unbiased estimator as

$$E[\hat{\theta}] = E\left[\sum_{i=1}^n x_i / n\right] = \frac{1}{n} \sum E[x_i]$$

we have shown $E[x_i] = \theta \quad \forall i \in [1, n]$ so $E[\hat{\theta}] = \frac{1}{n} \cdot n \cdot \theta$

$$\boxed{E[\hat{\theta}] = \theta}$$

2) Given $x_1, x_2, \dots, x_n$ are i.i.d

$$\text{We have } F_n(x) = \frac{1}{n} \sum_{i=1}^n I(x_i \leq x)$$

Required to show

$$E[F_n(x)] = F(x) \text{ \& } \text{Var}(F_n(x)) = \frac{1}{n} F(x)(1-F(x))$$

and $F_n(x) \rightarrow F(x)$ in probability as $n \rightarrow \infty$

$$\text{We have } E[I(x_i \leq x)] = \int_{-\infty}^x I(y \leq x) \cdot f dy + \int_x^{\infty} I(y \leq x) \cdot f dy$$

where f is probability density of x_i

$$\text{we also have } I(x_i \leq x) = \begin{cases} 1 & \text{if } x_i \leq x \\ 0 & \text{if } x_i > x \end{cases}$$

$$E[I(x_i \leq x)] = \int_{-\infty}^x 1 \cdot f \cdot dy + \int_x^{\infty} 0 \cdot f \cdot dy = \int_{-\infty}^x f dy$$

we have $\int_{-\infty}^x f dy = F(x)$ as $F(x)$ is C.D.F.

$$\begin{aligned} \text{So, } E[I(x_i \leq x)] &= F(x). \\ \Rightarrow E[F_n(x_i)] &= \frac{1}{n} \sum_{i=1}^n E[I(x_i \leq x)] \quad \text{as } x_1, \dots, x_n \text{ are i.i.d} \\ &= \frac{1}{n} (nF(x)) = F(x) \end{aligned}$$

$$\Rightarrow \boxed{E[F_n(x_i)] = F(x)}$$

$$\begin{aligned} \text{Var}(F_n(x)) &= E[F_n^2(x)] - (E[F_n(x)])^2 \\ &= \frac{1}{n^2} E\left[\sum_{i=1}^n (I(x_i \leq x))^2\right] - (F(x))^2 \\ &= \frac{1}{n^2} E\left[\sum_{i=1}^n (I(x_i \leq x))^2 + 2 \sum_{i=1}^n \sum_{j=i+1}^n I(x_i \leq x) \cdot I(x_j \leq x)\right] - (F(x))^2 \end{aligned}$$

We also have $E[I(x_i \leq x) \cdot I(x_j \leq x)] = E[I(x_i \leq x)] \cdot E[I(x_j \leq x)]$ for $i \neq j$
as $x_1, \dots, x_n$ are i.i.d

$$\begin{aligned} E[I(x_i \leq x) I(x_j \leq x)] &= (F(x))^2 \\ E[I(x_i \leq x)^2] &= \int_{-\infty}^x (I(y \leq x))^2 f(y) dy + \int_x^{\infty} (I(y \leq x))^2 f(y) dy \\ &= \int_{-\infty}^x (1)^2 f(y) dy = F(x). \end{aligned}$$

$$\begin{aligned} \text{Var}(F_n(x)) &= \left(\frac{1}{n^2}\right) \left[n \cdot F(x) + 2 \cdot \frac{n(n-1)}{2} \cdot (F(x))^2\right] - (F(x))^2 \\ &= \frac{1}{n} (F(x) + (n-1)F(x)^2) - F(x)^2 \\ &= \frac{1}{n} (F(x) - (F(x))^2) = \frac{1}{n} F(x)(1-F(x)) \\ \Rightarrow \boxed{\text{Var}(F_n(x)) = \frac{1}{n} F(x)(1-F(x))} \end{aligned}$$

For large n By law of large numbers we would

$$\text{have } \frac{F_n(x) - E[F_n(x)]}{\sqrt{\text{Var}(F_n(x))}} \rightarrow N(0,1).$$

$$\Rightarrow F_n(x) - F(x) \rightarrow N\left(0, \frac{1}{n} F(x)(1-F(x))\right)$$

as $F(x)$ is a C.D. $0 \leq F(x) \leq 1$

$$\text{so } \frac{1}{n} F(x)(1-F(x)) \rightarrow 0$$

as $n \rightarrow \infty$, $F_n(x) \rightarrow N\left(F(x), \frac{1}{n} F(x)(1-F(x))\right)$

as variance tends zero. value of $F_n(x)$ should tend to mean i.e. $\boxed{F_n(x) \rightarrow F(x)}$

1.2) Linear Regression

2) Given $\epsilon_1, \dots, \epsilon_n$ are i.i.d and $\epsilon_i \rightarrow N(0, \sigma_\epsilon^2)$
 $\forall i \in [1, n]$

Simple regression model

$$y_i = \alpha + \beta x_i + \epsilon_i$$

$$\Rightarrow y_i - \alpha - \beta x_i = \epsilon_i$$

$$y_i - \alpha - \beta x_i \sim N(0, \sigma_\epsilon^2)$$

$$y_i \sim N(\alpha + \beta x_i, \sigma_\epsilon^2)$$

Likelihood function $L(\alpha, \beta, \sigma_\epsilon^2)$

$$= f(x_1, x_2, \dots, x_n, \alpha, \beta, \sigma_\epsilon^2)$$

As $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ are Independent we would.

have $L(\alpha, \beta, \sigma_\epsilon^2) = f(x_1, \alpha, \beta, \sigma_\epsilon^2) \dots f(x_n, \alpha, \beta, \sigma_\epsilon^2)$

$$\begin{aligned} L(\alpha, \beta, \sigma_\epsilon^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma_\epsilon} e^{-\frac{(y_i - \alpha - \beta x_i)^2}{2\sigma_\epsilon^2}} \\ &= \left(\frac{1}{2\pi\sigma_\epsilon^2}\right)^{n/2} e^{-\frac{1}{2\sigma_\epsilon^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2} \end{aligned}$$

Log likelihood function becomes

$$\ln L(\alpha, \beta, \sigma_\epsilon^2) = -n/2 \log 2\pi \sigma_\epsilon^2 - (1/2\sigma_\epsilon^2) \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

For maximizing β we differentiate by β and equate to 0.

$$d \ln(\alpha, \beta, \sigma_\epsilon^2) / d\beta = 0 - (1/\sigma_\epsilon^2) \sum_{i=1}^n (y_i - \alpha - \beta x_i)(-x_i).$$

$$\sum x_i y_i - \alpha \sum x_i - \beta \sum x_i^2 = 0$$

Substituting $\bar{y} = \alpha + \beta \bar{x}$ i.e $\alpha = \bar{y} - \beta \bar{x}$ we get.

$$\sum x_i y_i - (\bar{y} - \beta \bar{x}) \sum x_i - \beta \sum x_i^2 = 0$$

$$\beta (\sum x_i^2 - \bar{x} \sum x_i) = \sum x_i y_i - \bar{y} \sum x_i$$

$$\beta = (\sum x_i y_i - \bar{y} \sum x_i - \bar{x} \sum y_i + n \bar{y} \bar{x}) / (\sum x_i^2 + n \bar{x}^2 - 2 \bar{x} \sum x_i)$$

$$\text{As } \bar{y} = \sum y_i / n \text{ \& } \bar{x} = \sum x_i / n \quad \bar{y} \sum x_i = \bar{x} \sum y_i = n \bar{y} \bar{x}$$

$$\hat{\beta} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

For $\hat{\alpha}$ we get $\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$ by substituting $\hat{\beta}$

(3) Given $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \epsilon$

For calculating sensitivity to only x_2 & x_3 we

take $\beta_2 = 1$ and $\beta_3 = 1$

So Restricted Regression equation becomes

$$y = \beta_1 + x_2 + x_3 + \beta_4 x_4 + \epsilon$$

and unrestricted is same as the original equation.

Given URSS = 397.2 and RRSS = 436.1

$$F\text{-test statistic} = \frac{RSS - URSS}{URSS} \times \frac{T-k-1}{m}$$

$$\text{Given } T = 144, k+1 = 4, m = 2$$

$$F\text{-test statistic} = \frac{436.1 - 397.2}{397.2} \times \frac{(144-4)}{2}$$

$$= (38.9 \times 140) / (397.2 \times 2)$$

$$\boxed{F\text{-test statistic} = 6.8555}$$

$H_0: \beta_2 = 1 \text{ and } \beta_3 = 1$ (null hypothesis)

$H_a: \beta_2 \neq 1 \text{ or } \beta_3 \neq 1$ (alternate hypothesis)

For 2-sided test, the F-dist. critical value at 0.005 significance level -

$$F_{2,140} = 5.504$$

Since calculated F-statistic (6.85) is greater than $F_{2,140}$ critical value

$\therefore$ Null hypothesis is rejected at 0.5% significance level

Hence $\beta_2 = 1$ & $\beta_3 = 1$ do not hold

1.3 Time Series Models

1. Y_t - AR(2) process
 $(Y_{t-k}) = \phi_1 (Y_{t-1} - \mu) + \phi_2 (Y_{t-2} - \mu) + \varepsilon_t$

② To prove: $\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2}$; $k > 0$

$$\rho_k = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)} \quad \left[\begin{array}{l} \text{since } \text{Var}(Y_t) = \\ \text{Var}(Y_{t-k}) \\ \text{Stationarity} \end{array} \right]$$

Let $V_k = \text{Cov}(Y_t, Y_{t-k})$

$$V_k = E[(Y_t - \mu)(Y_{t-k} - \mu)]$$

$$= E[(\phi_1 (Y_{t-1} - \mu) + \phi_2 (Y_{t-2} - \mu) + \varepsilon_t)(Y_{t-k} - \mu)]$$

$$= E[\phi_1 (Y_{t-1} - \mu)(Y_{t-k} - \mu)] +$$

$$E[\phi_2 (Y_{t-2} - \mu)(Y_{t-k} - \mu)] + E[\varepsilon_t (Y_{t-k} - \mu)]$$

$$= \phi_1 E[(Y_{t-1} - \mu)(Y_{t-k} - \mu)] + \phi_2 E[(Y_{t-2} - \mu)(Y_{t-k} - \mu)] + E[\varepsilon_t (Y_{t-k} - \mu)]$$

Now: $t-1$ and $t-k$ have lag of $k-1$
 $\therefore E[(Y_{t-1} - \mu)(Y_{t-k} - \mu)] = V_{k-1}$

$t-2$ and $t-k$ have lag of $k-2$
 $\therefore E[(Y_{t-2} - \mu)(Y_{t-k} - \mu)] = V_{k-2}$

$E[\varepsilon_t (y_{t-k} - \mu)] = 0$ since they are indep.

$$\therefore V_k = \phi_1 V_{k-1} + \phi_2 V_{k-2}$$

Divide both sides by $\text{Var}(y_t)$

$$\frac{V_k}{\text{Var}(y_t)} = \phi_1 \frac{V_{k-1}}{\text{Var}(y_t)} + \phi_2 \frac{V_{k-2}}{\text{Var}(y_t)}$$

$$f_k = \phi_1 f_{k-1} + \phi_2 f_{k-2}$$

hence proved

① Yule-Walker Equation -

$$f_k = \phi_1 f_{k-1} + \dots + \phi_p f_{k-p} ; k > 0$$

$$f_1 = \phi_1 f_0 + \phi_2 f_1 + \dots + \phi_p f_{p-1}$$

$$f_2 = \phi_1 f_1 + \phi_2 f_2 + \dots + \phi_p f_{p-2}$$

We can use these equations to estimate $\phi = (\phi_1, \dots, \phi_p)$ by using sample correlation f_i .

We can observe equations in ① are special case of Yule-Walker with $p = 2$

Use above equations and $f_0 = 1$

$$\begin{aligned} f_1 &= \phi_1 f_0 + \phi_2 f_{-1} \\ &= \phi_1 + \phi_2 f_1 \quad (\text{since } f_k = f_{-k}) \end{aligned}$$

$$\begin{aligned} f_2 &= \phi_1 f_1 + \phi_2 f_0 \\ &= \phi_1 f_1 + \phi_2 \end{aligned}$$

In matrix form:

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 1 & f_1 \\ f_1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix}$$

Hence proved

© Given $f_1 = 0.4$ and $f_2 = 0.2$

To find — ϕ_1 , ϕ_2 , f_3

Using equations from part © $\rightarrow$

$$f_1 = \phi_1 + \phi_2 f_1 \quad ; \quad f_2 = \phi_1 f_1 + \phi_2$$

$$0.4 = \phi_1 + \phi_2(0.4) \quad ; \quad 0.2 = \phi_1(0.4) + \phi_2$$

$$10 \phi_1 + 4 \phi_2 = 4 \quad ; \quad 4 \phi_1 + 10 \phi_2 = 2$$

— ① Multiply by 2.5

Subtract ① and ②:

$$10 \phi_1 + 25 \phi_2 = 5$$

— ②

$$21 \phi_2 = 1 \quad \Rightarrow \quad \phi_2 = \frac{1}{21} = 0.0476$$

$$10 \phi_1 = 4 - \frac{4}{21} = \frac{80}{21} \quad \Rightarrow \quad \phi_1 = \frac{8}{21} = 0.381$$

$$f_3 = \phi_1 f_2 + \phi_2 f_1$$

$$= (0.381) 0.2 + (0.0476) 0.4$$

$$= 0.0762 + 0.01904$$

$$= 0.095$$